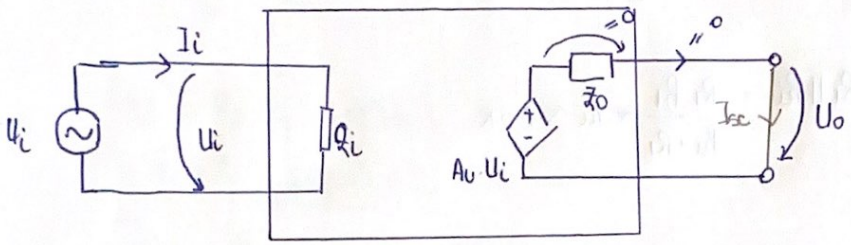
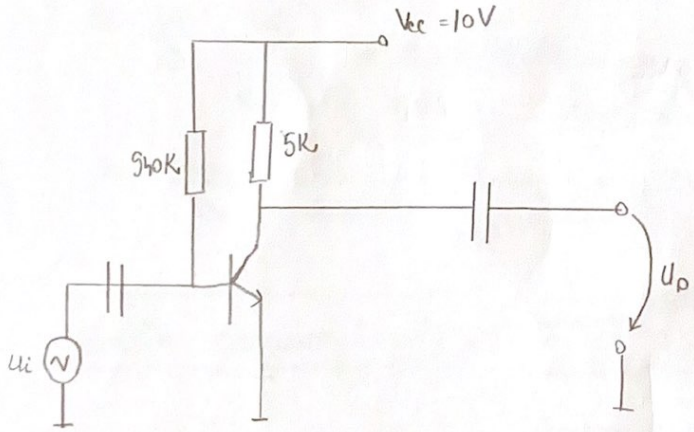
Curs 24 Noiembrie 2022

Amplificatoare



$$Z_i = \frac{U_i}{I_i} \qquad I_{sc} = \frac{A_U \cdot U_i}{Z_O}$$

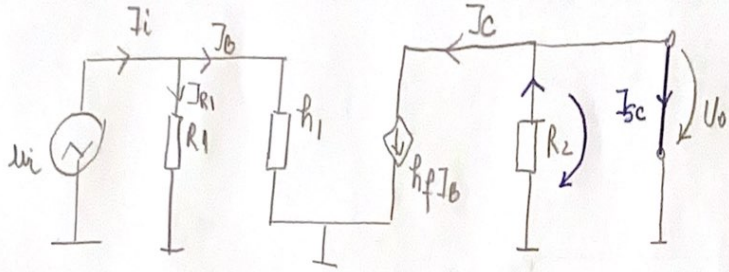
$$A_U = \frac{U_{OGol}}{U_i} \qquad \Rightarrow Z_O = \frac{A_u \cdot U_i}{I_{sc}}$$



$$\frac{I_c}{\beta} \cdot R_1 + U_{BE} = V_{cc} \Rightarrow I_c = 1mA$$

$$U_{CE} = V_{cc} - R_2 I_c = 5V > 0{,}6V \Rightarrow RAN$$

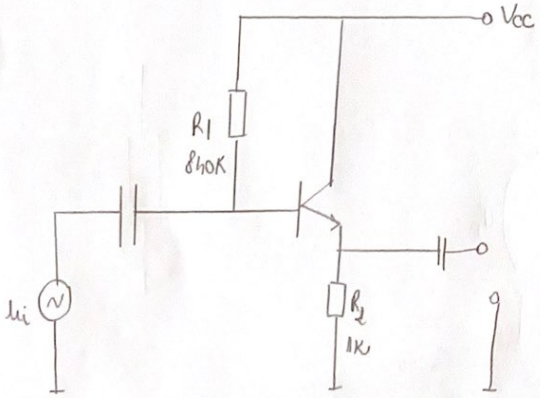
$$h_i = \frac{\beta}{40 I_c} \simeq 2{,}5\,K \qquad h_\beta = \beta = 100$$



$$Z_i = \frac{U_i}{I_i} = \frac{U_i}{\underbrace{\frac{U_i}{R_1}}_{I_{R1}} + \underbrace{\frac{U_i}{h_i}}_{I_B}} = R_1 \| h_i = \frac{R_1 \cdot h_i}{h_i + R_1} \simeq h_i \simeq 2.5K$$

$$A_U = \frac{U_{osol}}{U_i} = \frac{-I_C R_2}{U_i} = \frac{-\frac{U_i}{h_i} \cdot h_f \cdot R_2}{U_i} = -\frac{h_f}{h_i} \cdot R_2 = \frac{-100 \cdot 5K}{2.5K} = -200.$$

$$Z_o = \frac{A_U U_i}{I_{sc}} = \frac{A_U \cdot U_i}{-I_C} = \frac{-\frac{h_f}{h_i} \cdot U_i \cdot R}{-\frac{U_i}{h_i} \cdot h_f} = R_2$$

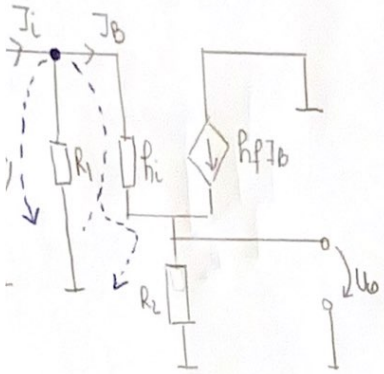


P.S.F

P.S.F. $I_{B1} R_1 + U_{BE} + I_E R_2 = V_{cc}$

$\Rightarrow I = 1mA \Rightarrow U_{CE} = 9V$

$h_i = \frac{\beta}{40 I_C} = 2.5K \quad h_f = \beta = 100$



$U_{GOL} = I_E \cdot R_2 = (h_f + 1) I_B R_2$

$I_B h_i + R_2 \cdot I_B (h_f + 1) = U_i$

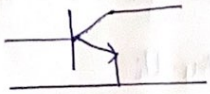
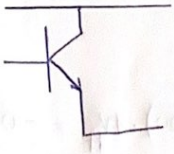
$$I_B = \frac{U_i}{h_i + R_2 (h_f + 1)} \simeq \frac{U_i}{R_2 \cdot h_f}$$

$$U_{GOL} = (h_f + 1) \frac{U_i}{R_2 h_f} \cdot R_2 \simeq U_i$$

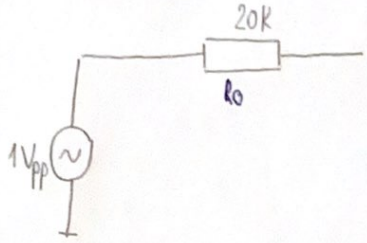
$$A_U = \frac{U_{GOL}}{U_i} = 1.$$

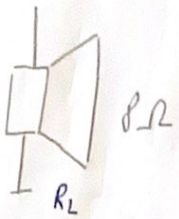
$$Z_i = \frac{U_i}{I_i} = \frac{U_i}{\frac{U_i}{R_1} + \frac{U_i}{R_2 h_f}} = R_1 \| R_2 h_f = \frac{840K \cdot 100K}{840K + 100K} \simeq 90K$$

$$Z_o = \frac{U_{GOL}}{I_{sc}} = \frac{U_i}{I_B (1 + h_f)} = \frac{U_i}{\frac{U_i}{R_2 h_f} (1 + h_f)} = \frac{U_i}{\frac{U_i}{h_i} (1 + h_f)} \simeq \frac{h_i}{h_f} \simeq 25\Omega$$

		A_i	Z_i	Z_o
Emitor comun (E.C)		-200	2,5K	5K
		A_i	Z_i	Z_o
Colector comun (C.C)		1	90K	25Ω

LINE OUT



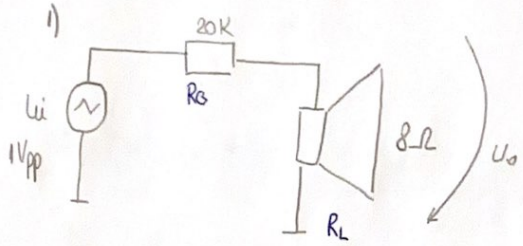


1) Conexiune directă

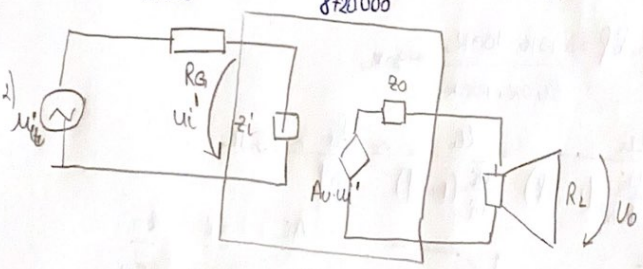
2) Emitor comun

3) Colector comun

1)



$$U_o = \frac{U_i}{20K+8\Omega} \cdot 8\Omega = 1 \cdot \frac{8}{8+20000} = 0,0003\ V_{pp} = 0,3\ mV_{pp}$$



$$U_o = \frac{R_L}{z_o + R_L} \cdot \frac{z_i}{R_G + z_i} \cdot A_U \cdot U_i$$

2) E.C.

$$U_o = \frac{8\Omega}{5K+8\Omega} \cdot \frac{2,5K}{20K+2,5K} \cdot (-200) \cdot 1V_{pp} = -0,035\ V_{pp}$$

3). C.C.

$$U_o = \frac{8\Omega}{25\Omega+8\Omega} \cdot \frac{90K}{90K+20K} \cdot 1 \cdot 1V_{pp} = 0,2\ V_{pp}$$

$U_0 = 0{,}0004\ V_{pp}$

$U_0 = -0{,}035 V_{pp}$

$U_0 = 0{,}2\ V_{pp}$.

$\frac{U^2}{R}$ potencia.